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Hydroacoustic survey standardization: Inter-vessel differences in fish densities and potential effects of vessel avoidance

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ABSTRACT

Hydroacoustics is used broadly to assess fish populations in marine and freshwater systems. Large-scale surveys often employ multiple vessels to complete a survey. Vessels can be a source of variation in multi-vessel surveys, and accounting for this variation is critical to precise and accurate assessments, whether as indices or measures of absolute abundance. We examined areal and volumetric density estimates, area backscattering coefficient (ABC), and mean target strength (TS) produced by three vessels in Lake Erie along three cross-lake transects in July 2014. Our study showed positive correlation in areal ABC, mean TS, and density among vessels; however, there was an almost 30 % higher density obtained with the largest vessel, on average. As a result, vessel choice could contribute to variation and/or bias in forage fish surveys deploying multi-vessels conducted on Lake Erie. Additionally, in some situations, differential depth patterns of fish contributed to vessel related differences in survey results suggesting a difference in vessel avoidance behaviors among targeted forage species, primarily emerald shiner *Notropis atherinoides* and rainbow smelt *Osmerus mordax*. This work demonstrates the potential magnitude of differences among vessels and highlights the need to identify and account for or otherwise control for these effects when monitoring or managing fisheries through standardized hydroacoustic assessments surveys.

1. Introduction

Hydroacoustics has been a cornerstone of fisheries assessments for decades in marine and freshwater systems (Simmonds and MacLennan, 2005; Rudstam et al., 2012). It is relied on for stock assessments (e.g., Rose, 2003; Yule et al., 2009) and for assessing prey fish populations to guide predator stocking (e.g., Warner et al., 2008). Great progress has been made over time to understand sources of uncertainty (e.g., Simmonds and MacLennan, 2005) and to standardize data collection methods (Parker-Stetter et al., 2009; CEN, 2014; ICES, 2014; Draštík et al., 2017), resulting in improved accuracy, precision, and defensibility of hydroacoustic assessments. In contrast, comparatively little work has been conducted on comparisons among vessels when multiple vessels are used in a single survey, including the potential for systematic and differential vessel avoidance behaviors by target species.

Broad scale hydroacoustic surveys are frequently conducted with more than one vessel. Rose (2003) reported three different vessels used in 1995 to survey Atlantic cod *Gadus morhua* near Newfoundland, Canada. Eight different vessels are listed by the International Council for the Exploration of the Sea (ICES, 2014) for fisheries assessments in the Baltic Sea. On the Swedish Lakes Vättern and Vänern, three different vessels ranging in length from 12–30 m have been used since 1992 for annual surveys and research (T. Axenrot and A. Sandström, Swedish University of Agricultural Sciences, pers. comm., 4 August 2017). Since the 1990s, multiple vessels ranging in length from 6.7 m to 55 m have been used, sometimes during the same survey, in the five Laurentian Great Lakes to assess yellow perch *Perca flavescens* (Roberts et al., 2012), cisco *Coregonus artedi* (Myers et al., 2015), prey fish, mysids, and zooplankton communities (FTG, 2017; Warner et al., 2009, 2008; Watkins et al., 2015; Stockwell et al., 2007; Holbrook et al., 2006). In

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most cases when multiple vessels are used, details on vessel size or noise produced by vessels are not reported, even though these variables may cause differences in estimated fish densities or differential avoidance responses from the target species. Without understanding the relative differences between vessels used within a survey, the impact on accuracy and precision of that survey cannot be assessed.

Vessel avoidance by fishes is well studied and documented. Both diving and lateral movements have been reported in response to passing trawl and hydroacoustic vessels (Olsen, 1990; Ona and Godø, 1990; Mitson, 1995; Draštík and Kubečka, 2005; Wheeland and Rose 2014; reviews in De Robertis and Handegard, 2013; DuFour et al., 2018; Trumpickas et al., 2020), and likely includes both behaviors to some extent (Soria et al., 1996; Misund, 1997). Diving can alter orientation angle (Ona et al., 2007; Hjellvik et al., 2008) and compress the swim bladder, both mechanisms that affect target strength (De Robertis and Handegard, 2013) and subsequently density estimates when using echo integration (De Robertis and Wilson, 2011). Lateral movement can result in fish not being ensonified (De Robertis and Handegard, 2013), hence potentially biasing density estimates low (DuFour et al., 2018). Vessel avoidance behaviors are driven by auditory and visual cues (e.g., vessel noise, vessel light, etc.) as well as proximity to the vessels themselves (Misund, 1997); however, a myriad of other factors may also contribute (e.g., species, size, water temperatures, etc.) and as a result avoidance patterns are not always predictable. For example, De Robertis and Wilson (2011) reported vessel-related differences in a survey of walleye pollock *Gadus chalcogrammus*, where the quieter of two similarly sized vessels, one of which was specifically designed to meet noise reduction standards, had higher total backscatter. Conversely, Ona et al. (2007) reported less avoidance of a smaller vessel despite the larger vessel being noise-reduced and quieter. This suggests that even a well-informed understanding of vessel avoidance behaviors and their general impact on acoustic measurements should not preclude exploring inter-vessel differences within surveys.

The need or desire to use multiple vessels within a survey, or change vessels through time, is driven primarily by logistics and safety concerns (FTG 2017), and therefore eliminating multiple vessel surveys is not a realistic option. An implicit assumption of multi-vessel surveys is that hydroacoustic data among them are comparable, or stated in terms of avoidance, that fish avoidance reactions are similar among vessels. If avoidance differs among vessels and it is not known how, then results between vessels are not comparable, regardless of the efforts made to standardize data collection and analysis protocols (see Hjellvik et al., 2008). Additionally, this vessel-specific measurement related bias is then incorporated into overall survey estimates as additional spatially related variation (Larsen et al., 2001), which could impact the ability to make management relevant decisions (Wagner et al., 2009). This is of course true for any active fishing gear in which fish behavior may be affected by the vessel, such as electrofishing and trawling. Ideally, a measure of differences in avoidance or bias would be available when multiple vessels are used in a single survey, so that corrective actions could be taken during the survey planning or data analysis stages.

Here we compare hydroacoustic estimates collected during a standard survey from three research vessels spanning the range of vessel sizes (7.6 m, 21 m, and 55 m) used for hydroacoustic assessments on Lake Erie, one of the Laurentian Great Lakes of North America. We sought to 1) assess differences in areal estimates including density (fish/ha), area backscattering coefficient (ABC), and *in situ* target strength (TS) among vessels, as well as 2) depth related differences among the same metrics using volumetric density (fish/10,000 m^3), to identify systematic or differential biases and infer potential fish avoidance behaviors relative to each vessel. These results directly inform hydroacoustic survey design and implementation on Lake Erie, and more broadly consider the impact of variation among vessels in multi-vessel surveys.

2. Methods

2.1. Target species and system

The Lake Erie central basin hydroacoustic survey has implemented consistent methodology since 2004 targeting rainbow smelt *Osmerus mordax* and emerald shiner *Notropis atherinoides* (FTG 2017). The full survey design includes eight cross-basin transects, a total of 643 km, accompanied by midwater trawls which allow apportionment by species. This survey is conducted annually during the new moon in July to increase availability of target species, which at the time of survey range between 25 and 130 mm (Rudstam et al., 2003). Yearling and older rainbow smelt, mean TS ranging between -55 and -45 dB (Rudstam et al., 2003), reside near bottom during daylight and migrate vertically into the water column at night (Appenzeller and Leggett, 1995) making them more available to the survey. Conversely, a mixture of young-of-year rainbow smelt and emerald shiner, mean TS ranging between −62 and −55 dB (Rudstam et al., 2003), are positively phototactic and more likely to be higher in the water column during daylight, thus, more available to the survey at night. These two species, residing in high density aggregations along the thermocline, typically compose >90 % of the overall fish biomass captured in midwater trawl samples conducted concurrently with hydroacoustic sampling since 2004 (e.g., FTG 2017). Vessel avoidance effects are a concern in this system, as the central basin is relatively shallow, with sampled depths ranging between 15–25 m, resulting in the potential interaction between targeted fishes and survey vessels. Additionally, multiple vessels (6 since 2004 including those in this study) within and across years have collected data using multiple deployment methods (e.g., thru-hull tubes, tow body, and fixed pole mounts). As a result, the Lake Erie central basin hydroacoustic survey, like many others, has likely been impacted by vessel related bias and variation.

2.2. Survey description

Hydroacoustic data were collected from three different vessels along three cross-lake transects in central Lake Erie 21 July – 23 July 2014 (Fig. 1). The three vessels spanned the entire range of vessel sizes currently and historically used on Lake Erie: R/V North River (7.6 m), R/V Muskie (21 m), and R/V Lake Guardian (55 m; Table 1). The R/V Muskie has collected hydroacoustic data since being placed in service in 2012 and is considered the standard vessel to which results of the other two vessels are compared. The cross-lake transects were a standard part of the Lake Erie central basin hydroacoustic survey starting and ending at the 15-m depth contour and varying slightly in length (Transect 1 = 69 km, Transect 2 = 72 km, and Transect 3 = 69 km). One randomly chosen vessel collected data on the standard survey transect, and the other two vessels were randomly assigned to either starboard or port of the standard transect vessel. This resulted in the R/V Muskie designated as the on-transect vessel for transect 1 (R/V Lake Guardian to starboard, R/V North River to port), the RV Lake Guardian on transect for transect 2 (R/V Musky starboard, R/V North River port), and R/V North River on transect for transect 3 (R/V Musky starboard, R/V Lake Guardian port). Vessels maintained a 500-m spacing to reduce potential for noise from one vessel affecting another vessel and to accommodate operational needs of the R/V Lake Guardian. Lake Erie's central basin surveys were conducted at night, during the new moon phase following the Great Lakes Standard Operating Procedure (GL-SOP; Parker-Stetter et al., 2009).

We used a side-by-side survey design rather than follow-the-leader for several reasons. First, Lake Erie's central basin is relatively shallow (≤25 m), and a high-density proportion of the fish biomass resides in the epilimnion just above the thermocline; therefore, vessel effects are expected. In this scenario, a follow-the-leader method would compound vessel effects producing vessel relative density estimates along the sample path. Additionally, annual monitoring has shown fish abundance

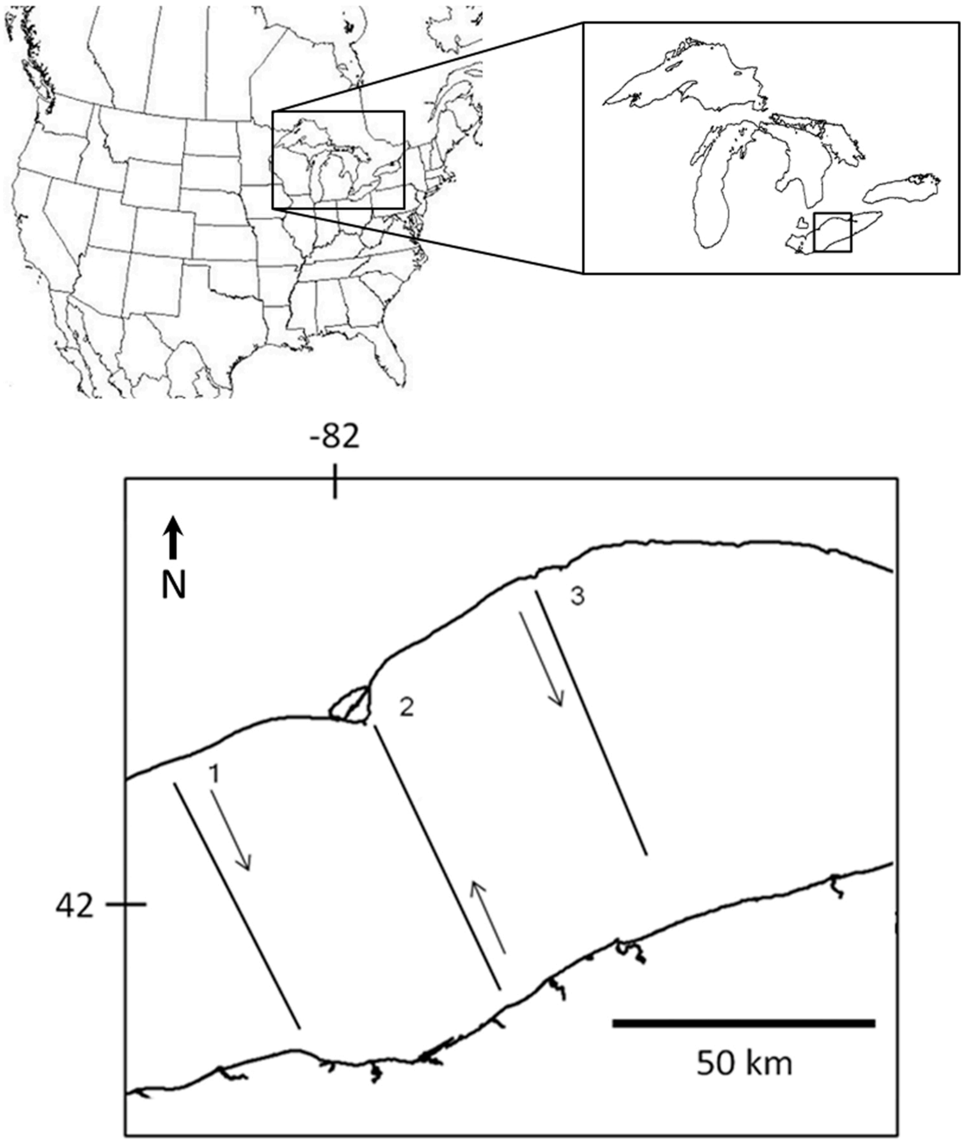

Fig. 1. Transects sampled using hydroacoustics in central Lake Erie in 2014.

Table 1
Specifications of vessels used for hydroacoustic data collection and transducer beam angle and deployment depth in the central basin of Lake Erie.

Vessel	Length (m)	Beam (m)	Draft (m)	Speed (knots)	6-dB beam angle (°)	Mount (depth)
R/V North River	7.6	2.6	1.2	5.7	7.60	Towed BioFin (1.2 m)
R/V Muskie	21	5.5	1.7	5.0 – 5.1	7.90	Through-hull tube (1.7 m)
R/V Lake Guardian	55	12	3.4	5.1	7.60	Towed BioFin (2 m)

in Lake Erie to be variable, but highly correlated over ranges less than 2 km. Therefore, we would expect ships to encounter similar densities along each sample interval and that the collection of intervals would represent the overall mean density along the sample transect. Finally, the side-by-side survey design produces a data set that feeds naturally into routine statistical methods (i.e., ANOVA).

2.3. Data collection

Each vessel collected data using a 120-kHz BioSonics split-beam transducers with DT-X surface units operating at 4 pin s^{-1} and with a 0.0004 s pulse duration (per the GL-SOP). Transducers were deployed on a towed body (BioSonics BioFin) from the R/V Lake Guardian and R/V North River, and inside a through-hull tube on the R/V Muskie. The R/V Lake Guardian transducer was equipped with a pitch and roll sensor to monitor orientation during collection. Vessel speeds ranged between 5.0 and 5.7 knots. Data collection began no earlier than 30 min after sunset and ended no later than 30 min before sunrise. Each transducer was independently calibrated prior to data collection by the manufacturer or using the standard sphere methods (Foote et al., 1987). Acoustic offsets were applied, as needed (R/V North River = −0.38 dB, R/V Muskie = −0.13, and R/V Lake Guardian = not needed), during data processing.

2.4. Data processing

Data were processed using Echoview (version 6.1 and version 7.0, Echoview, 2016) including 2-km elementary distance sampling units (EDSU) and 1-m depth layers beginning at 5 m from the surface.

Although the nearfield for the transducers used was < 2 m, the 5-m threshold was used to ensure the first layer evaluated was beneath the draft of the largest vessel. Single echo detections (SED) were filtered using Echoview Method 2 for split-beam data using -68 dB compensated TS threshold, 6-dB pulse length determination level, 0.6 minimum and 1.5 maximum normalized pulse length, BioSonics beam compensation model, 6-dB maximum beam compensation, and 3 for maximum standard deviation of minor- and major-axis angles. Following the GL-SOP, we established a minimum TS for fish targets of -68 dB (Rudstam et al., 2003; Kočovský et al., 2013), then set a range-dependent TS-based volume backscattering coefficient (S_v) threshold 6 dB lower than minimum TS to remove back scattering from smaller targets (Rudstam et al., 2009). Ambient noise estimates were determined for each vessel by integrating below the first bottom echo, and noise estimates were subtracted from S_v (Parker-Stetter et al., 2009). We exported three hydroacoustic measurements including ABC, backscattering cross-section (σ_{bs}), and mean TS.

2.5. Density calculations

For each vessel, transect, EDSU, and layer we calculated volumetric estimates of density (fish/10,000 m^3) from ABC and σ_{bs} following standard equations (Simmonds and MacLennan, 2005; Parker-Stetter et al., 2009); density (fish/10,000 m^3) = (ABC/σ_{bs})/10,000. A small proportion of cells did not record S_v larger than the post-processing threshold (-74 dB) applied in Echoview resulting in 0 values for ABC; therefore, we added the smallest recorded ABC value to all cell specific ABC records across vessels to circumvent analysis issues using 0 data, such as taking the log of 0. Additionally, there is the potential for high single-target densities in this survey (Rudstam et al., 2003), which could lead to failure of the single target detection algorithms causing the acceptance of multiple echoes as single targets. This can lead to biased cell-specific σ_{bs} and mean TS estimates, where mean TS = 10*$\log_{10}(\sigma_{bs})$, which in turn can bias density estimates. To avoid this bias, we calculated the Nv index (Sawada et al., 1993), which identifies the density at which multiple echo contamination is likely, for each cell using the detected *in-situ* σ_{bs}. If Nv values exceeded 0.1, we used mean σ_{bs} of cells with Nv < 0.1 from the same layer, transect, and vessel to re-scale ABC (i.e., density (fish/10,000 m^3) = (ABC/mean σ_{bs})/10,000) and replace biased *in situ* mean TS (i.e., mean TS = 10*$\log_{10}$(mean σ_{bs})). We summarized the occurrence of biased TS estimates by calculating the frequency of Nv values > 0.1 across the survey, by vessel, by transect and vessel, and by layer.

2.6. Statistical analyses

We were interested primarily in differences between vessels potentially caused by lateral or vertical vessel avoidance. We used two different approaches to explore differences between vessels including correlation and two different mixed-effect ANOVA models. Model parameter estimates were then used to calculate marginal posterior differences (MPDs) which are analogous to the R statistic described in Kieser et al., 1987 (see De Robertis et al., 2008 and Trumpickas et al., 2020 for additional applications) and allow for proportional comparisons among factors levels. For each approach, we performed six analyses comparing the R/V North River vs. the R/V Muskie and the R/V Lake Guardian vs. R/V Muskie for the response variables areal density (fish/ha) or volumetric density (fish/10,000 m^3), ABC (x10^{-7}), and mean TS (dB). We ran separate analyses for each vessel comparison because the R/V North River only completed 1.4 transects and vessel-specific intervals along transect were misaligned due to slightly varying start and stop locations. As such, including all vessels' data into a single analysis for each approach and measurement type would create unbalanced and misaligned datasets leading to biased results. Running separate analyses allowed us to maximize the use of available comparison data. Exploring the different hydroacoustic measurements (areal and volumetric density, ABC, and mean TS) allowed us to evaluate each component contributing to density and abundance estimates and identify whether there were systematic or differential biases between research vessels. For example, areal and volumetric density are the primary measurement of interest, but these are calculated from the ABC and σ_{bs} measurements (i.e., density (fish/m^3) = ABC/σ_{bs},), where mean TS is directly related to mean σ_{bs} (i.e., $\sigma_{bs} = 10^{TS/10}$). By analyzing these separately, we can determine how each component differs by vessel and make inferences on potential behavioral responses that may impart systematic or differential patterns. Therefore, evaluating each component with variable specific models for each comparison alloweds us to thoroughly investigate the potential for vessel related biases.

We analyzed the data in two different forms to explore differences among vessels (areal estimates) as well as depth related differences (volumetric estimates) using correlation and AVOVA (specific details below). To accommodate each of the individual variables using ANOVA models, we used different distributional assumptions and model structures. The volumetric density (fish/10,000 m^3), and ABC (x10^{-7}) variables were log-normally distributed; therefore, we log-transformed (ln) these variables to approximate normality prior to fitting the ANOVA models. The mean TS variable is on a $\log_{10}$ scale, and additional transformation was not necessary to approximate normality prior to model fitting. We fit all ANOVA models in R (R Core Team, 2019) using *lme4* package (Bates et al., 2015) for mixed-effects models and determined the significance of main and interaction effects using the *anova* function from the *lmerTest* package (Kuznetsova et al., 2017), which produced a Type III ANOVA table including p-values for F-tests using the Satterthwaite's method for determining denominator degrees of freedom. Next, we drew random samples from the parameter estimates of each model using *Stan* (Stan Development Team, 2019), called through the *brms* package (Bürkner, 2017) to provide more flexibility with plotting and estimate comparisons. *Stan* is a modeling program that uses Hamiltonian Monte-Carlo (HMC) algorithms to efficiently sample parameter spaces. Each model used 4 chains, and between 1,000 and 3, 000 warm-up and sampling iterations, depending on convergence criteria and effective sample size estimates, with no thinning. We assessed model convergence by visually inspecting chain histories for autocorrelation within chains and divergent transitions, and the $\widehat{R}$ statistic, where convergence was assumed when $\widehat{R}$ values on all model parameters were $\leq$ 1.1 (Gelman et al., 2014).

2.6.1. Areal analyses

Initially, we performed correlation using layer-aggregated estimates to determine how whole-water-column estimates compared between vessels. We first summed the layer-specific estimates of volumetric fish density (fish/10,000 m^3) to produce an areal density estimate (fish/ha) for each EDSU. Similarly, we summed the ABC (x10^{-7}) values for each 1-m layer to obtain an ABC value for the whole water column for each EDSU. We used the average of mean TS (dB). Mean TS per interval was calculated by first averaging layer-specific σ_{bs} measurements on the linear scale, and then transforming to the log-linear scale where, mean TS = 10*$\log_{10}(\sigma_{bs})$. Aggregating the data in this way produced paired estimates of density, ABC, and mean TS for each EDSU across the surveys. We used Pearson's correlation coefficient (ρ) to determine the strength of association between the paired measurements for each vessel comparison and measurement (6 total). Correlations were carried out in R (R Core Team, 2019) using the *cor* function in the *stats* package.

Next, we compared the magnitude of differences for each areal estimate between paired vessels using one-way, random-effects ANOVAs, where vessel was the fixed effect. This survey occurred over a large spatial scale, potentially introducing non-vessel related variation into the data set; therefore, we included EDSU as a random effect to account for this source of variation. Additionally, using the vessel-specific marginal posterior estimates, including uncertainty, we calculated the MPD between vessels for each measurement (areal density, ABC, and mean

TS) on the scale of interest. MPDs were calculated by subtracting the individually correlated values of the jointly distributed HMC samples within each factor level. For example, in the R/V North River vs. R/V Muskie comparison, the Vessel factor (β) includes two levels β_{NR} and β_{M}, respectively. Each of the level-specific marginal posterior estimates are composed of i jointly distributed samples. By subtracting each jointly distributed sample from one another, we produced an estimate of the difference with associated uncertainty. Differences on the natural-log scale, due to prior transformation, were returned to the original scale by exponentiation ($MPD_{vessel} = e^{\beta_{NR_i} - \beta_{M_i}}$) and represent proportional differences. However, mean TS, a derivative of σ_{bs} ($TS = 10*\log_{10}(\sigma_{bs})$), was modeled on the log-10 scale and was converted back to the linear scale for comparison ($MPD_{vessel} = 10^{(\beta_{NR_i} - \beta_{M_i})/10}$). All MPDs were calculated relative to the standard vessel (R/V Muskie); positive value indicated a larger estimate for the comparison vessel while negative values indicated larger estimates for the Muskie.

2.6.2. Volumetric analyses

In addition, we evaluated the influence of depth-related variation on the vessel differences using volumetric data and two-way, random-effects ANOVAS with an interaction term. Similar to the one-way ANOVA described above, we included EDSU as a random effect to account for spatially related variation in the data. Additionally, we included fixed vessel and depth layer as main effects, and the two-way interaction between vessel and depth layer. A significant layer main effect was anticipated due to previously observed partitioning of rainbow smelt year classes (FTG 2017) and diel vertical migration of rainbow smelt (Appenzeller and Leggett, 1995). The primary responses of interest for assessing avoidance were the vessel main effect and the vessel*depth-layer interaction, which is the vessel-specific effect for a given depth-layer. A significant interaction indicates that differences in densities between vessels depend on the depth of the layer, which indicates vessel-induced avoidance.

3. Results

3.1. Survey and data collection

All three vessels completed the first transect (Fig. 1). On the second night, rough seas forced the R/V North River to abandon data collection roughly half-way through the second transect, but the R/V Muskie and R/V Lake Guardian carried on. Continuation of rough seas during the third night prevented the R/V North River from collecting additional data. Although the smaller vessel could not continue, the weather was like typical conditions experienced by the standard vessel R/V Muskie. The final sample sizes for correlation and ANOVA were 1.4 transects for the R/V Muskie and R/V North River comparison and 3 transects for the R/V Muskie and R/V Lake Guardian comparison. Start and stop locations among the three vessels over the first 1.4 transects were slightly offset which resulted in slightly different spatial coverages between comparisons, causing median areal densities (fish/ha) to differ between comparisons (Table 2).

3.2. Density estimate adjustments

A small percentage of cells in each comparison survey (5/1466 in the R/V North River vs. R/V Muskie and 6/3208 in the R/V Lake Guardian vs. R/V Muskie) had S_v values below the applied threshold, therefore ABC values were 0. The Nv index was generally low (i.e., < 0.1), suggesting that σ_{bs} and mean TS estimates were largely unbiased. The overall mean percentage of high (> 0.1) index values was 4.9 %. The R/V North River had the lowest percentage of high Nv values (mean 3.5 %, range 2.0 %–7.7 %), while the R/V Lake Guardian was slightly higher (mean 4.9 %, range 0.6 %–7.4 %) and the R/V Muskie was the highest (mean 6.4 %, range 1.7 %–9.5 %). Transect 2 had the highest mean percentage of high Nv at 7.4 % for the R/V Lake Guardian, 7.7 % for the R/V North River, and 9.5 % for the R/V Muskie. The percentage of high Nv values was 5 % or lower in layers 5–11 and ranged from 5.9 to 13.4 % in layers 12−18. The highest Nv was in layer 16 (13.4 %).

3.3. Areal analysis results

Hydroacoustic measurements varied within (north to south) and across transects (West to East). In the R/V North River vs. R/V Muskie comparisons, areal density (fish/ha), and ABC were variable with no general trend across Transect 1 (north to south), while mean TS increased slightly (Fig. 2). Moving east to Transect 2 (south to north), areal density and ABC were slightly higher, while mean TS in Transect 2 were lower and decreased south to north. The patterns in areal density, ABC, and mean TS in the R/V Lake Guardian vs. R/V Muskie (Fig. 3) were like the previous comparison for Transects 1 and the beginning of Transect 2. Additionally, all three variables increased over the second half of Transect 2. These values increased or remained relatively high at the beginning of Transect 3 (north) and declined through the end of Transect 3 (south). Results from all vessels indicate an increasing trend in areal density progressing eastward (Table 2). ABC and mean TS increased across transects West to East and South to North for transects 2 and 3, but only mean TS increased from north to south along Transect 1 (Figs. 2 and 3). Correlations between vessels were generally high for fish/ha, and ABC (≥ 0.79; Table 3, Figs. 2 and 3), but lower (≤ 0.48) for mean TS. In all cases, correlations were higher between R/V Lake Guardian and R/V Muskie than between R/V North River and R/V Muskie.

The difference in areal estimates between R/V North River and R/V Muskie were not significant for all variables (Table 3). Areal density (fish/ha) and mean TS were similar between vessels as indicated by large p-values as well as MPD values close to 1 (Fig. 4). Although ABC was not statistically significant, measures were proportionally lower (0.90) with very little overlap in MPD values with 1, indicating the R/V North River consistently recorded lower ABC than the R/V Muskie. Areal comparisons between R/V Lake Guardian and R/V Muskie were significant for all variables except mean TS (Table 3). Areal density (fish/ha) and ABC were both proportionally larger for the R/V Lake Guardian (Fig. 4). MPD values suggest areal density and ABC were higher by 30 % and 15 % respectively, relative to the R/V Muskie. Mean TS values were similar between R/V Lake Guardian and R/V Muskie.

Table 2
Median areal densities, # of ESDU (N) and 10 & 90 percentiles in parenthesis, (fish/ha) between 5 m and the bottom along three surveyed transects in this study.

Vessel	Transect 1	Transect 2		Transect 3
		First part	Second part	
Lake Guardian	7480, N = 32, (3957−13445)	8924, N = 14, (7033−18267)	17360, N = 23, (5440−37692)	15459, N = 35, (7384−34626)
Muskie	5495, N = 32, (2850−10175)	7390, N = 14, (5454−14917)	11939, N = 23, (3757−25787)	12382, N = 35, (5256−22330)
North River	5912, N = 35, (2918−9963)	8548, N = 14, (5490−11543)		
Muskie	5869, N = 35, (2981−9945)	8068, N = 14, (5903−14595)		

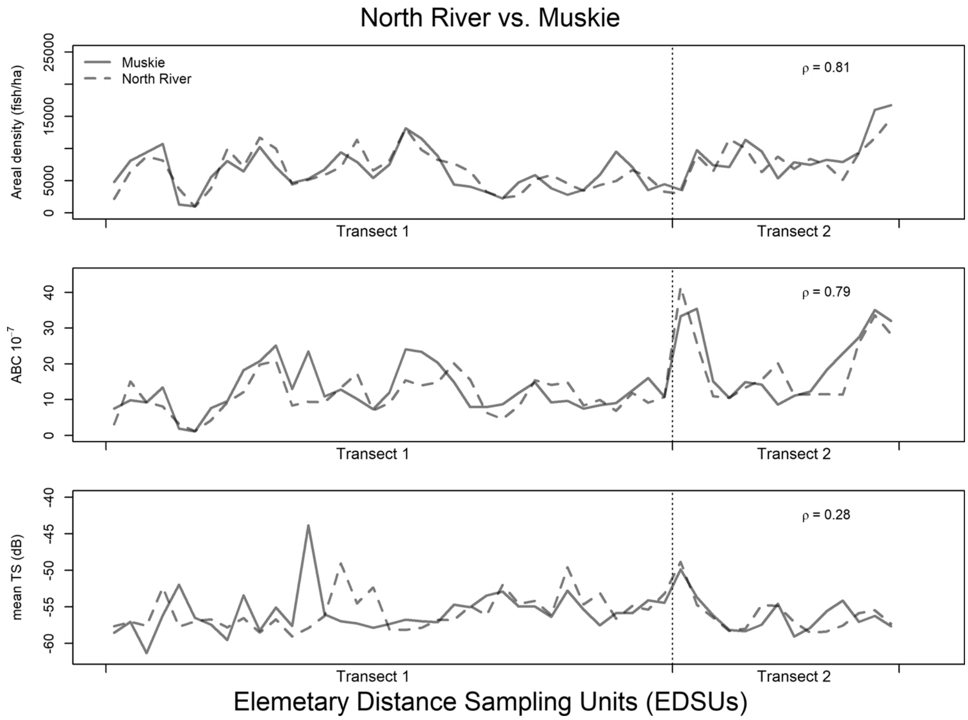

Fig. 2. Raw data values for elementary distance sampling units including areal density (fish/ha) and ABC (x10^{-7}), mean TS (dB), and SEDs (#/10,000 m^3) from the R/V North River (dashed lines) and R/V Muskie (solid lines).

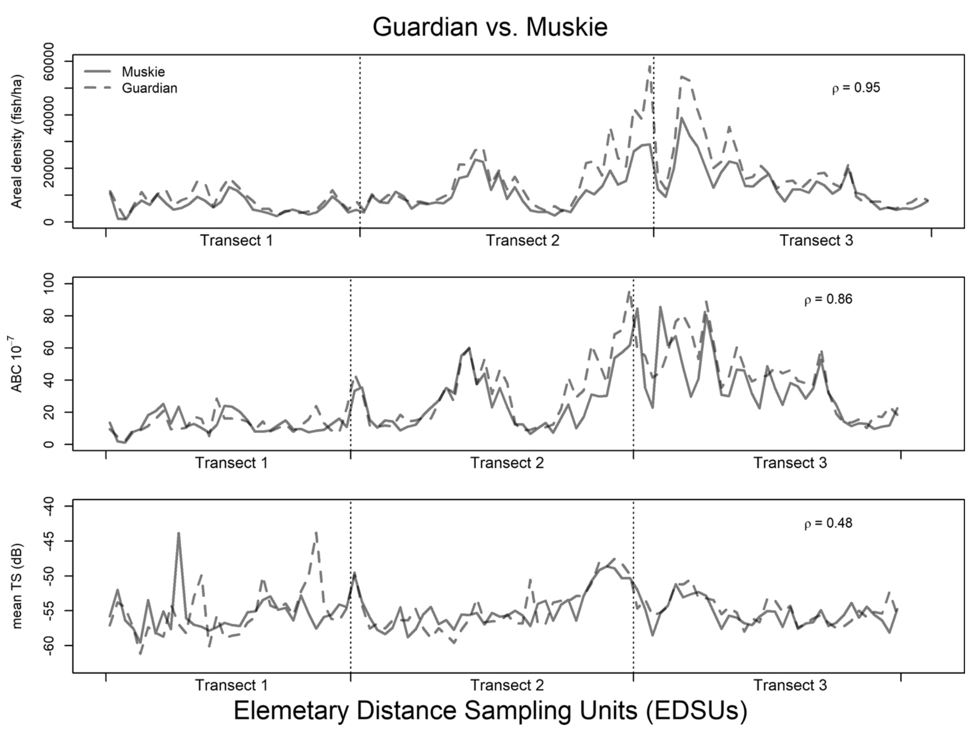

Fig. 3. Raw data values for elementary distance sampling units including areal density (fish/ha), ABC (x10^{-7}), mean TS (dB), and SEDs (#/10,000 m3) from the R/V Lake Guardian (dashed lines) and R/V Muskie (solid lines).

3.4. Volumetric analysis results

For the R/V North River vs. R/V Muskie depth-related comparison, the layer main effect was significant for all variables as expected due to vertical layering of fish abundance in Lake Erie. The vessel main effect was significant for all variables except mean TS (Table 4). The vessel-by-layer interaction term was significant for mean TS but not density or ABC, indicating layer values for mean TS depended on the vessel. For the R/V Lake Guardian vs. R/V Muskie comparison, the same pattern was observed for the layer and vessel main effects, but the interaction term was not significant for any of the variables (Table 5).

Differences in marginal means and MPDs for the four response

Table 3
ANOVA results including mean square error (Mean Sq), numerator degrees of freedom (Num DF), denominator degrees of freedom (Den DF), F statistic (F value), and p-value (Pr(>F)) for the between reference vessel (R/V Muskie) and comparison vessels (R/V North River and R/V Lake Guardian) analyses using areal density, area backscattering coefficient (ABC), and mean target strength (mean TS). Additionally, the correlation values (Cor (ρ)) and marginal proportional differences (MPD) between vessel specific estimates are included.

Value	Mean Sq	Num DF	Den DF	F value	Pr(>F)	MPD *	Cor (ρ)
North River vs. Muskie							
Areal density (fish/ha)	0.0040	1	48	0.0674	0.796	0.99	0.81
ABC $x10^{-7}$	0.2786	1	48	3.7553	0.059	0.90	0.79
mean TS (dB)	0.0354	1	48	0.0072	0.933	1.00	0.28
Lake Guardian vs. Muskie							
Areal density (fish/ha)	3.6524	1	103	117.900	**<0.001**	1.30	0.95
ABC $x10^{-7}$	0.9805	1	103	12.964	**<0.001**	1.15	0.86
mean TS (dB)	0.0050	1	103	0.001	0.973	1.00	0.48

* < 1 indicates lower values for North River or Lake Guardian.

variables varied by inter-vessel comparison. Mean ABC was slightly lower for the R/V North River compared to the R/V Muskie in nearly all acoustic layers (Fig. 5). Mean TS increased from surface to bottom for the R/V North River but varied without trend for the R/V Muskie. Mean TS was lower for the R/V North River than the R/V Muskie through 12 m and higher below 17 m. The net result on estimated volumetric density was marginally higher densities for the R/V North River through 10 m and increasingly lower densities from 10 to 21 m. Results were quite different for the R/V Lake Guardian vs. R/V Muskie comparison. Mean ABC for the R/V Lake Guardian was higher than that of the R/V Muskie in nearly all acoustic layers (Fig. 6). Mean TS for the R/V Lake Guardian was similar to or smaller than the R/V Muskie through 17 m, then was larger through 21 m. The net result was that estimated volumetric density for the R/V Lake Guardian was higher in nearly all layers.

4. Discussion

Hydroacoustic vessel comparison studies in the central basin of Lake Erie demonstrated positive correlation in areal ABC, *in situ* mean TS, and density of fish sampled among vessels; however, some differences in magnitudes of fish sampled were identified including an almost 30 % higher density obtained with the largest vessel. As a result, vessel choice could contribute to variation and/or bias in forage fish surveys conducted on Lake Erie deploying multi-vessels. A deeper look into the influence of depth, using volumetric measurements (i.e., 1-m depth layers), indicated that in some situations differential depth patterns contributed to vessel related differences in survey results. This suggests a difference in vessel avoidance behaviors among targeted forage species, primarily emerald shiner and rainbow smelt. Due to Lake Erie's shallow depths (15–25 m) and known depth stratification of the fish community, where yearling and older rainbow smelt are in the hypolimnion and young-of-year rainbow smelt and emerald shiner are in the epilimnion (Rudstam et al., 2003), different vessels would, thus, produce different measures of fish community composition. Conversely, some vessel-related differences in survey results appeared to be consistent with depth which suggests either consistent depth-related vessel avoidance behaviors among target species and/or systematic differences in measurements from hydroacoustic systems (i.e., measurement error or deployment bias). In situations where multiple vessels are used to complete a full survey, differential or systematic bias can contribute to overall uncertainty in survey estimates and should be mitigated during

Table 4
ANOVA results for the between reference vessel (R/V Muskie) and comparison vessel (R/V North River) analyses using volumetric density, area backscattering coefficient (ABC), and mean target strength (mean TS).

Factor	Mean Sq	Num DF	Den DF	F value	Pr(>F)
fish/10,000 m^3					
Vessel	32.3	1	1386.5	12.5	**< 0.005**
Layer	221.2	15	1388.0	85.6	**< 0.005**
Vessel:Layer	3.2	15	1386.5	1.2	0.231
ABC $x10^{-7}$					
Vessel	30.2	1	1386.2	9.0	**< 0.005**
Layer	187.7	15	1387.8	55.8	**< 0.005**
Vessel:Layer	1.1	15	1386.2	0.3	0.992
mean TS (dB)					
Vessel	0.7	1	1382.9	0.1	0.822
Layer	40.9	15	1385.2	3.1	**< 0.005**
Vessel:Layer	26.2	15	1382.9	2.0	**0.015**

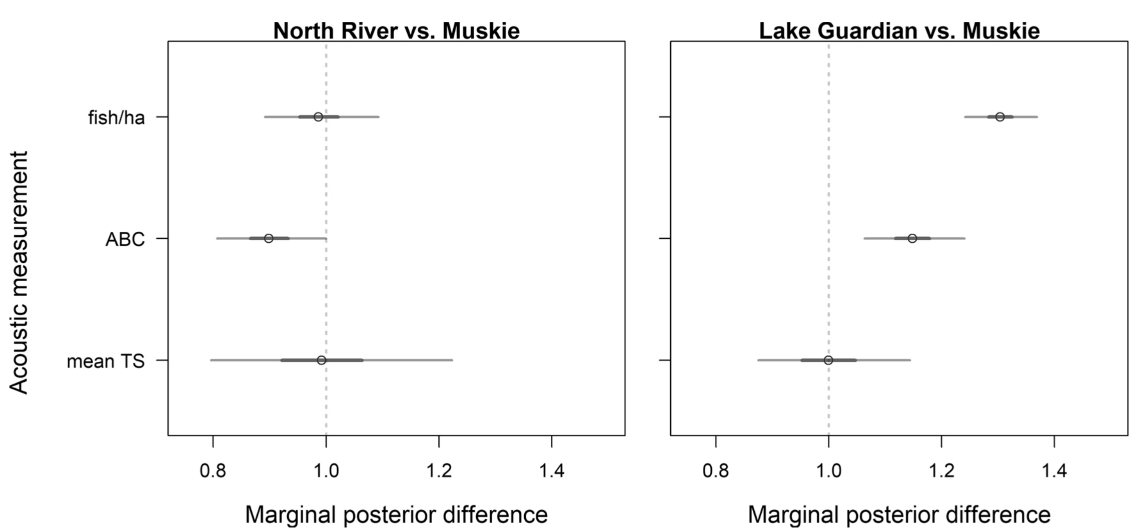

Fig. 4. Marginal posterior differences proportional to the R/V Muskie using areal data for density (fish/ha), ABC, mean TS, and SEDs. Circles represent the mean estimate and dark and light gray bars represent 50 and 95 % credible intervals. Values overlapping 1 are not significantly different and correspond to ANOVA results in Table 3.

Table 5
ANOVA results for the between reference vessel (R/V Muskie) and comparison vessel (R/V Lake Guardian) analyses using volumetric density, area backscattering coefficient (ABC), and mean target strength (mean TS).

Factor	Mean Sq	Num DF	Den DF	F value	Pr(>F)
fish/10,000 m^3					
Vessel	44.4	1	3073.6	21.8	**< 0.005**
Layer	429.6	15	3075.0	210.4	**< 0.005**
Vessel:Layer	0.5	15	3073.6	0.2	0.999
ABC x10^{-7}					
Vessel	30.3	1	3073.5	10.8	**< 0.005**
Layer	313.7	15	3074.8	111.8	**< 0.005**
Vessel:Layer	0.9	15	3073.5	0.3	0.995
mean TS (dB)					
Vessel	25.6	1	3071.9	1.8	0.182
Layer	235.1	15	3074.1	16.4	**< 0.005**
Vessel:Layer	21.0	15	3071.9	1.5	0.108

the survey design phase or corrected for after collection. These problems are not unique to hydroacoustic surveys on Lake Erie or within the Laurentian Great Lakes, as multiple vessels are used in other broad-scale surveys in both marine and freshwater settings (e.g., Rose, 2003; ICES, 2014).

Areal hydroacoustic measurements, where areal density (fish/ha) per EDSU is the basic data obtained from a hydroacoustic survey, suggest that the R/V North River and R/V Muskie produced similar estimates of overall areal density (fish/ha), whereas the R/V Lake Guardian produced consistently higher estimates. Proportional differences between R/V North River and R/V Muskie were smaller than between R/V Lake Guardian and R/V Muskie. These results are similar to Trumpickas et al. (2020), who reported no difference in overall nautical area scattering coefficient (NASC - directly proportional to ABC used here) collected in a section of Georgian Bay, Lake Huron, from two vessels similar in size to those used here. In contrast, the R/V Lake Guardian indicated larger differences than the R/V Muskie where areal density (fish/ha) and ABC were proportionally higher (1.30 and 1.15, respectively). These results were contrary to expectations, as presumably the larger R/V Lake Guardian would increase vessel related avoidance leading to reduced density estimates (Ona et al., 2007; De Robertis and Handegard, 2013). In any case, from a multi-vessel survey perspective, coupling the R/V North River and R/V Muskie would limit inter-vessel related variation while including the R/V Lake Guardian would contribute an additional source of variability to the overall forage densities.

Some surveys partition acoustic returns by species, and these species are not evenly distributed within the water column. Our analyses of volumetric measurements (i.e., including 1-m depth layers) suggests that differential avoidance between the R/V North River vs. R/V Muskie occurred throughout the water column, but the near surface differences are of primary importance because this is where most of the biomass is located. The R/V North River encountered slightly higher densities in the upper two layers (6 – 7 m), with similar densities between 8–10 m, and slightly lower densities from 11 m and below. This finding is consistent with Trumpickas et al. (2020) who indicated a lower mean depth of NASC from the larger vessel, relative to the smaller vessel, suggesting less near surface acoustic return from the larger vessel. The similarity between studies suggests that the shift in ABC and density from near surface to lower layers could be caused by vertical avoidance (i.e., diving). In addition, lower TS in the 6−11 m layers for the R/V North River, a smaller vessel, is consistent with greater lateral avoidance of the R/V Muskie, an intermediate sized vessel. Lateral-avoidance bias is likely greater closer to the ship and will decrease with distance from the ship because of the increasing volume of the acoustic cone with depth (see Soria et al., 1996). DuFour et al. (2018) reported similar differential avoidance by larger single targets of the R/V Muskie and a vessel of identical hull design, propulsion, and transducer deployment as the R/V North River. This finding may be supported here, as the smaller vessel (R/V North River) recorded higher mean TS near bottom – an area inhabited by larger benthically oriented fishes. In the case of near surface horizontal avoidance, present fish would not be ensonified leading to underestimated densities of young-of-year rainbow smelt and emerald shiner in the upper layers from larger vessels. In the case of vertical avoidance (i.e., diving), if substantial proportion of fish from the epilimnion migrated into the hypolimnion then this acoustic return could be mis-apportioned to other species or age groups, simultaneously resulting in underestimation of upper layer fish (i.e., young-of-year rainbow smelt and emerald shiner) and overestimation of lower layer fish (i.e., adult rainbow smelt). As such, understanding differential avoidance patterns among vessels within a single survey would help with strategic deployment to minimize impacts.

The R/V Lake Guardian vs. R/V Muskie comparison suggests that there were significant vessel differences in volumetric density (fish/10,000 m^3), and ABC; however, these differences were not depth dependent. Additionally, ABC and volumetric density (fish/ha) were systematically lower for the R/V Muskie compared to the R/V Lake Guardian resulting in the net outcome of ∼24 % higher density estimates across layers for the R/V Guardian, the larger vessel, than for the R/V Muskie, an intermediate sized vessel. Consistency in mean TS, and depth consistent differences in ABC and volumetric density (fish/10,000 m^3) suggest either systematic avoidance behaviors with depth or inherent differences in measurements collected from different hydroacoustic systems. Hydroacoustic system performance was not expected to be a major source of measurement error in our survey, as all systems were calibrated by the manufacturer or using the standard sphere method (Foote et al., 1987) prior to this survey, and minor system specific mean TS offsets were applied as needed during post processing. It is possible that deployment method contributed to the observed differences. The R/V Lake Guardian used a towed BioFin deployed from a boom that extended ∼ 10 m from the vessel center, while the R/V Muskie used a through-hull tube that collected within a moon-pool at the vessel center. The R/V Lake Guardian transducer was equipped with a pitch and roll sensor which indicated some roll movement and a nose-down orientation (∼ -6°) across the sampled transects (Table A1). We assume the R/V Muskie transducer, while subject to pitch and roll during high wave conditions, was the most stable deployment during this survey; although, its transducer was not equipped with a pitch and roll sensor. We would expect that differences contributed from deployment method stability would be most evident in the TS measurements as fish orientation and accurate off-axis single target positioning estimates within the beam are critical for accurately compensated TS estimates. Even so, the mean TS estimates between the two vessels, although variable across intervals and with depth, were not different on average across the entire survey. Additionally, the R/V North River also used a towed BioFin, which was presumably (no pitch and roll sensor) subjected to similar transducer movement conditions as the R/V Lake Guardian, but this did not result in systematic differences in mean TS compared to the R/V Muskie. It is possible that lateral avoidance, along with vessel induced displacement, funneled fish along the side of the R/V Lake Guardian increasing the density under the extended towed BioFin, a pattern that has been observed in marine systems using multi-beam sonar (Soria et al., 1996). In any case, these systematic differences can contribute inter-vessel related variation to the results of multi-vessel surveys and should be identified prior to establishing survey logistics.

Our results for the R/V North River vs. R/V Muskie comparison are consistent with most of the literature on vessel avoidance (Misund, 1997; De Robertis and Handegard, 2013; Trumpickas et al., 2020), including one study that reported greater avoidance of a larger vessel despite that vessel having been specifically designed to emit less noise (Ona et al., 2007) compared to a smaller and louder vessel. As demonstrated by Trumpickas et al. (2020), although overall areal density

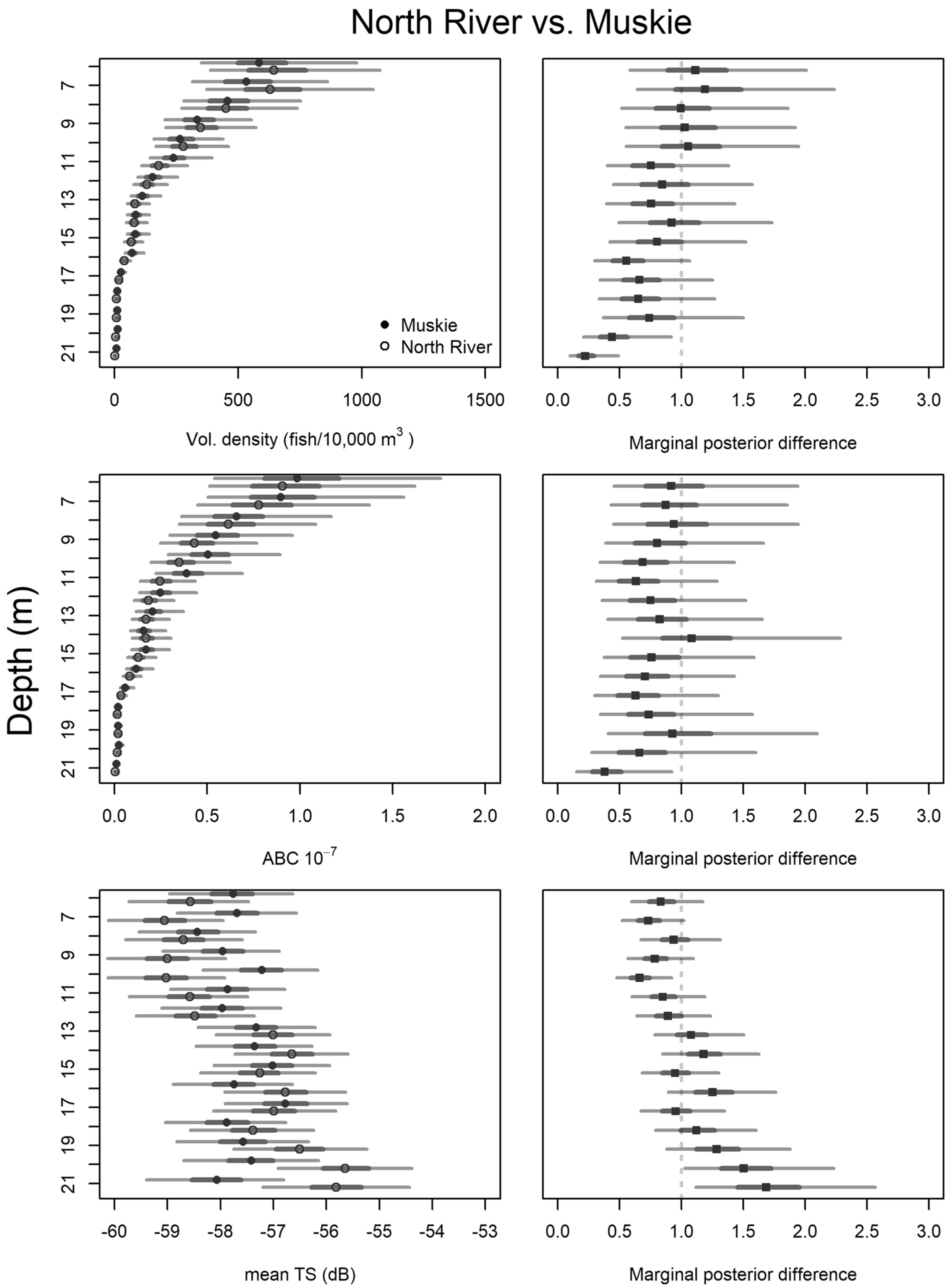

Fig. 5. Marginal median estimates for vessel main effects from volumetric density (fish/10,000 m^3), ABC (x10^{-7}), mean TS (dB), and SEDs (#/10,000 m^3) ANOVA models for R/V North River vs. R/V Muskie comparison and associated marginal posterior differences. Dark gray bars represent 50 % credible intervals, while light gray bars representing 95 % credible intervals.

estimates may not have shown vessel related differences, depth related differences may be present suggesting differential vessel related avoidance behaviors. Curiously, the R/V Lake Guardian vs. R/V Muskie did not follow the same pattern as the larger vessel recorded higher densities, and differences appeared to be consistent with depth. It is at first difficult to explain this contrast, as we expect that visual avoidance or attraction is unlikely because surveys were conducted at night, and vessel lighting was minimized during data collection (see Misund, 1997 for marine examples). However, we did not directly measure frequency-specific noise produced by the vessels used in this study (De Robertis and Handegard, 2013), or the effects of bow waves or many other potential causes summarized by Mitson (1995), some of which

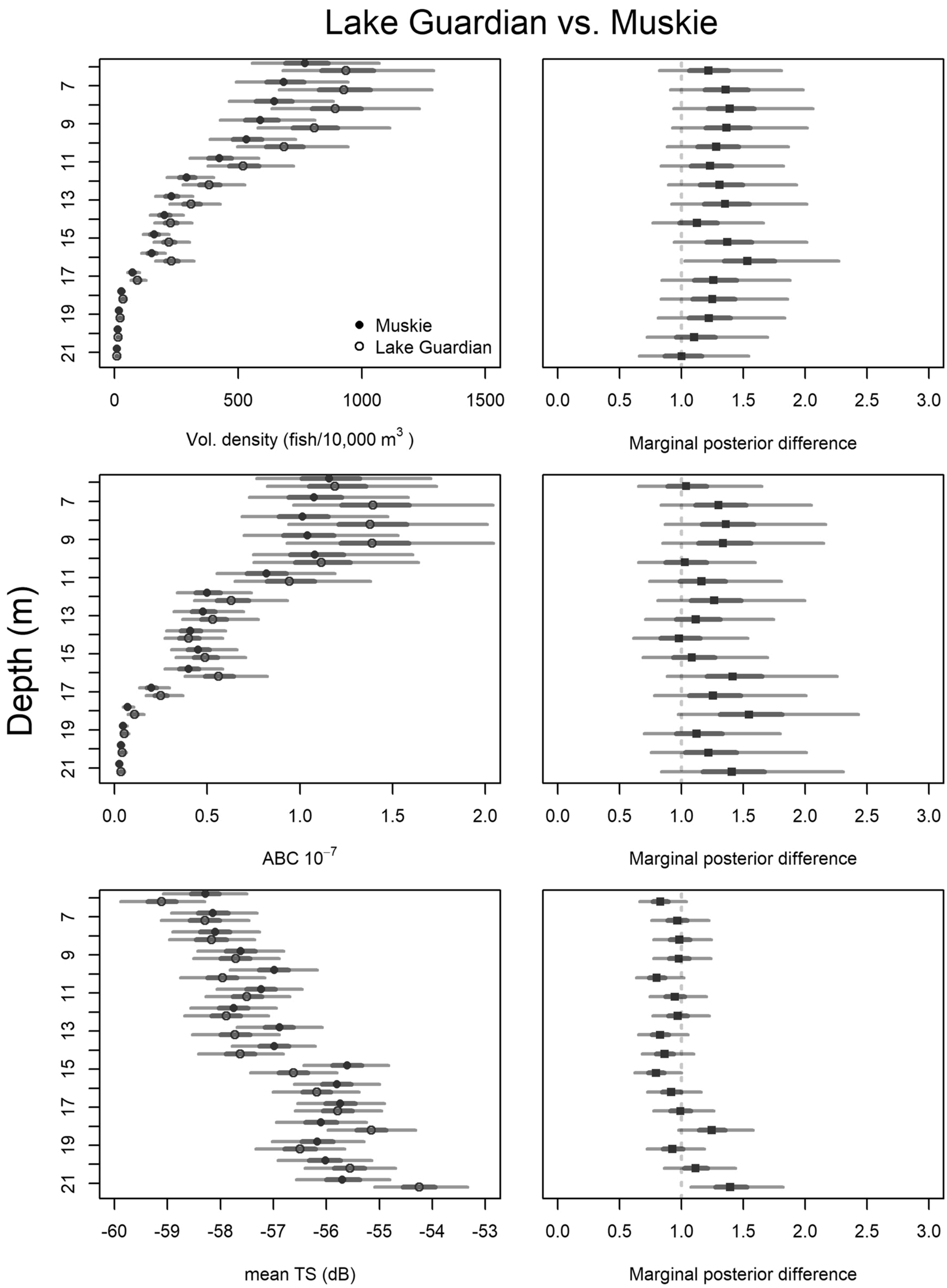

Fig. 6. Marginal median estimates for vessel main effects from volumetric density (fish/10,000 m^3), ABC (x10^{-7}), mean TS (dB), and SEDs (#/10,000 m^3) ANOVA models for R/V Lake Guardian vs. R/V Muskie comparison and associated marginal posterior differences. Dark gray bars represent 50 % credible intervals, while light gray bars representing 95 % credible intervals.

may have contributed to avoidance. We did further explore deployment methods and found that the R/V Lake Guardian towed BioFin collected at slightly nose-down orientation and included some role movement, while the through-hull deployed R/V Muskie transducer was likely stable. Although it does not appear differences in transducer orientation and motion contributed to measurement related biases, as mean TS estimates were similar between vessels, the position of the transducer relative to the vessel may have had an effect, as avoidance behaviors could lead to higher densities along vessel sides relative to directly under a vessel (Soria et al., 1996). These differences further highlight the importance of comparing vessels deployed within a single survey, as the deployment methods are limited by vessel construction and as seen here, could substantially influence density estimates within a survey.

Avoidance and the effects on fish density estimates are well studied (review in De Robertis and Handegard, 2013); however, differential vessel avoidance adds another layer of potential bias and variation to hydroacoustic density estimates generated from surveys using multiple vessels. For example, depth related differences in mean TS and distributions can differentially affect species and size-class abundance estimates, depending on the magnitude of differences and apportionment method used (Yule et al., 2013). Additionally, if the fish community reacts differently to vessels across a survey, then this measurement related bias is incorporated into the overall survey estimates as additional spatially related variation (Larsen et al., 2001; Wagner et al., 2009). Current assessment programs for forage fish species in the Laurentian Great Lakes is a case in point. In all lakes, surveys combine hydroacoustic estimates of overall pelagic fish density with trawl-captured fish to apportion hydroacoustic densities to species and age classes within layers and EDSU. Apportionment method varies by lake (see Yule et al., 2013), but all methods rely to some degree on *in situ* TS, which can be biased by avoidance behaviors. In Lake Erie, three different vessels (the R/V Muskie used in this study is one of them), ranging in size from 7.6–21 m, are used to collect hydroacoustic data, and three others between 12.8–17.7 m, are used to sample fishes with midwater trawls in the epilimnion, metalimnion, and hypolimnion as defined by temperature and dissolved oxygen profiles (FTG, 2017). In Lake Michigan, at least four vessels between 17.4 and 30.5 m are used (Warner et al., 2008). In Lake Ontario, at least 3 different vessels are used (Holden et al., 2017). Moving forward, hydroacoustic surveys and standard operating procedures in the Laurentian Great Lakes (Parker-Stetter et al., 2009) as well as other broad-scale surveys using multiple vessels (e.g., Rose, 2003; ICES, 2014), may need to address the impact of differential and systematic vessel avoidance relative to management and monitoring objectives (Wagner et al., 2013).

Post-data-collection alternatives for applying correction factors when multiple vessels are used have also been attempted. Hjellvik et al. (2008) developed methods of applying correction factors to account for avoidance. One of their methods failed to adequately account for diving herring (Clupea harengus), and the other may cause errors in density estimates above and below the layer of highest abundance. Most importantly, their observed avoidance response was not constant across multiple experiments, demonstrating that it does not suffice to use a single "correction factor" across surveys. An array of active sonar stations similar to those used for assessing vessel avoidance (De Robertis and Handegard, 2013) that are deployed within all sampling strata and that are passed by all vessels used in an assessment could provide survey- and vessel-specific correction factors. Paired sampling of autonomous or towed vehicles and survey vessels (De Robertis and Handegard, 2013; Riha et al., 2017; Grow et al., 2020) would provide continuous corrections analogous to post-processing removal of background noise (De Robertis and Higginbottom, 2007) and potentially better account for surface-oriented fish. However, any methods that use stationary listening stations or paired autonomous vehicles in addition to the vessel survey increases overall survey cost by increasing hardware required to conduct a survey as well as the additional time for post-processing and analysis. A more radical option that could greatly decrease overall survey cost over the longer term would be to forego altogether using large vessels that cause fish to scatter and rely solely on autonomous vehicles (Greene et al., 2014).

4.1. Recommendations

We recommend continued research (e.g. De Robertis and Handegard, 2013; ICES, 2014; Grow et al., 2020) on inter-vessel differences as well as alternative methods for minimizing and accounting for these differences. In addition, establishing clearly defined management and survey objectives (Wagner et al., 2013) prior to survey design and implementation will help identify and minimize potential bias prior to data collection. However, if managers or researchers desire absolute abundance estimates, the reduction of vessel avoidance in general as well as quantification of inter-vessel differences will provide the most accurate measures of stock abundances. Finally, to minimize the impact on hydroacoustic time series and as more information becomes available, we suggest that standard operating procedures (e.g., Parker-Stetter et al., 2009; CEN, 2014) be updated to specifically address measures to reduce or otherwise account for vessel avoidance effects and inter-vessel variation.

CRediT authorship contribution statement

Mark DuFour: application-coding, model development, writing, reviewing, and editing. **Patrick Kocovsky:** conceptualize, survey design, data collection/processing, writing and editing. **John Deller:** survey design, data collection/processing, writing and editing. **Paul Simonin:** survey design, data collection/processing, writing and editing. **Lars Rudstam:** supervision, conceptualize, survey design, writing and editing

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.fishres.2021.105948.

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